

Benchmarking AutoML and Agentic Pipelines Against Expert-Tuned Models on Small Datasets

A Literature-Grounded Review and Benchmarking Framework for Small-Data Tabular Machine Learning

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Small tabular datasets create a difficult machine-learning regime because limited observations increase estimator variance, amplify sensitivity to preprocessing and hyperparameter choices, and make conventional train/validation/test partitioning statistically fragile. At the same time, automated machine learning (AutoML) has evolved from automated hyperparameter optimization toward systems capable of model selection, feature preprocessing, ensembling, and increasingly autonomous pipeline construction. Large language model (LLM)-based agentic systems introduce a further paradigm in which machine-learning workflows can be planned, implemented, executed, evaluated, and revised by autonomous or multi-agent systems. This review synthesizes literature on conventional AutoML, small-data-specific models, and agentic machine-learning systems, with emphasis on the methodological conditions required for a fair comparison with expert-tuned models. The review organizes the literature around four methodological families: expert-tuned pipelines, conventional AutoML, LLM-driven agentic pipelines, and specialized small-data foundation models. Across these families, predictive performance alone is insufficient for comparison. Validation design, search-space definition, computational cost, stability, calibration, reproducibility, interpretability, and human effort materially affect conclusions. The central gap is the limited availability of controlled studies that compare strong expert baselines, conventional AutoML, agentic systems, and small-data-specific models under the same small-data protocol. A literature-grounded benchmark is therefore proposed using strict information boundaries, nested validation, controlled compute budgets, repeated resampling, and multi-dimensional evaluation.

Additional Key Words and Phrases: Automated Machine Learning, AutoML, Agentic AI, LLM Agents, Small Data, Tabular Machine Learning, Hyperparameter Optimization, Benchmarking, Expert Tuning, TabPFN

1 Introduction

Machine learning practice traditionally combines algorithmic methods with human decisions about preprocessing, feature engineering, validation, model families, hyperparameters, error analysis, and final model selection. Automated machine learning (AutoML) emerged partly to reduce this manual burden by automating model selection, hyperparameter optimization, preprocessing, feature selection, and pipeline construction [? ?]. The resulting systems changed the role of the practitioner from repeatedly implementing individual experiments toward defining search spaces, budgets, and evaluation protocols.

Early AutoML systems framed development as a structured optimization problem. Auto-sklearn combined Bayesian optimization, meta-learning, and ensemble construction [?], while TPOT used genetic programming to search over machine-learning pipelines [?]. Later methods introduced resource-aware search, including Hyperband [?] and BOHB [?], while FLAML emphasized efficient search under constrained budgets [?]. AutoGluon-Tabular emphasized model diversity, stacking, and multi-layer ensembles [?].

The small-data setting changes the methodological question. When the number of observations is limited, a validation split consumes a relatively large fraction of available information. Repeatedly evaluating many configurations on the same validation data can also make the selection process sensitive to validation noise. Thus, the ability to run more experiments does not itself create more

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statistical information. A fair comparison must separate the optimization procedure from the statistical reliability of the evidence used to choose a model.

The question becomes more complex with large language models (LLMs). LLM-based agents can inspect data, generate code, interpret errors, propose preprocessing, execute experiments, and revise their own workflow. AutoML-Agent, for example, uses multiple specialized agents, retrieval-augmented planning, and multi-stage verification for full-pipeline AutoML [?]. This expands the optimization target from a predefined configuration space toward workflow design itself.

At the same time, specialized small-data models challenge the assumption that extensive search is always necessary. TabPFN uses a pretrained Transformer for small tabular classification and reports competitive performance with very low tuning cost [?]. These developments motivate a review that does not assume a universal winner. Instead, the relevant question is under what data, budget, validation, and information-access conditions each methodological family is useful.

This paper makes four review-level contributions. First, it develops a taxonomy of four methodological families. Second, it synthesizes their strengths and limitations using common evaluation dimensions. Third, it identifies methodological threats that can make comparisons misleading, especially validation overfitting, search-space bias, leakage, nondeterminism, and unequal computational budgets. Fourth, it proposes a controlled benchmark protocol for comparing expert-tuned, conventional AutoML, agentic, and specialized small-data approaches.

2 Review Scope and Methodological Framing

The reviewed literature is organized around a common question: how should automated and expert-driven approaches be compared when the available tabular dataset is small? The source literature covers foundational AutoML, hyperparameter optimization, benchmarking, small-data-specific modeling, and LLM-based agentic automation [? ? ? ? ? ? ? ?].

For this review, “small data” is treated as an operational research regime rather than a rhetorical label. A practical starting criterion is

$$n_{\text{train}} \leq 1000, \quad (1)$$

while recognizing that sample size alone does not characterize difficulty. Feature dimensionality, class imbalance, missingness, noise, and feature-to-observation ratio can substantially alter the statistical problem.

A useful descriptor is

$$\rho = \frac{p}{n}, \quad (2)$$

where p is the number of features and n is the number of training observations. Equation 2 captures an important distinction: two datasets with the same number of observations may have very different levels of dimensional pressure.

2.1 Review Dimensions

The literature is synthesized using eight dimensions:

- (1) predictive performance;
- (2) robustness and stability under resampling;
- (3) computational and monetary cost;
- (4) calibration and statistical validity;
- (5) interpretability and auditability;
- (6) pipeline complexity;
- (7) reproducibility; and
- (8) human effort.

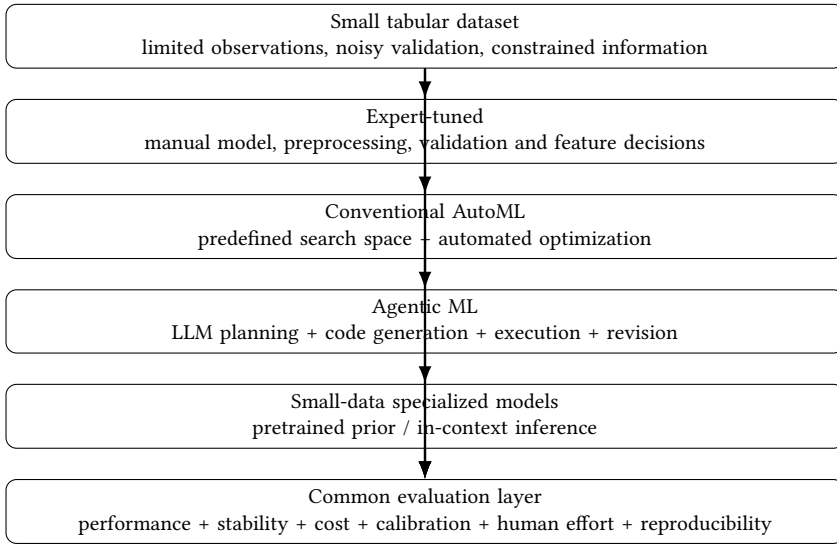


Fig. 1. Literature-grounded taxonomy for comparing methodological families in small-data tabular machine learning.

This multidimensional framing follows the methodological concerns emphasized by AutoML benchmarking work [?] and by the small-data and agentic literature reviewed in the source paper.

3 Taxonomy of Approaches

Figure 1 organizes the literature into four methodological families. The taxonomy is analytical rather than a claim that the categories are mutually exclusive in implementation.

3.1 Expert-Tuned Models

Expert tuning uses human knowledge to select preprocessing, candidate algorithms, hyperparameter ranges, validation procedures, feature engineering, and final models. The advantage is that domain knowledge can be applied even when it is difficult to encode in a generic search space. An expert can, for example, recognize when a high-capacity model is inappropriate for the sample size or when class imbalance makes raw accuracy misleading.

The limitation is that expert search is constrained by time, experience, preferences, and human bias. A competent expert may therefore explore a narrower region of the hypothesis space than a computational search system. The correct comparison is consequently not “expert versus automation” in the abstract, but a comparison under explicitly documented expertise, time, and information budgets.

3.2 Conventional AutoML

Conventional AutoML automates search over a predefined configuration space. Its main methodological advantage is systematic exploration and repeatability. Auto-sklearn combines Bayesian optimization, meta-learning, and ensemble construction [?]; TPOT uses genetic programming for pipeline optimization [?]; Hyperband allocates resources adaptively and terminates weak configurations early [?]; BOHB combines Bayesian optimization with bandit-based resource allocation [?]; and FLAML emphasizes efficient search under limited budgets [?].

Table 1. Comparative synthesis of methodological families

Family	Core mechanism	Strengths identified in the literature	Main limitations for small-data comparison
Expert-tuned	Human selection of models, preprocessing, validation and tuning	Domain knowledge; flexible decisions; can encode problem-specific constraints	Human time; bias; limited search coverage; expertise is difficult to standardize
Conventional AutoML	Automated search over pre-defined configurations and pipelines	Systematic search; repeatability; scalable evaluation; reduced manual optimization	Search-space bias; validation overfitting; compute cost; performance depends on benchmark protocol
Agentic ML	LLM planning, code generation, execution, interpretation and iterative revision	Semantic task reasoning; flexible workflow design; broad automation	Nondeterminism; leakage risk; hidden optimization loops; reproducibility and API/tool dependence
Small-data specialized	Pretrained prior and in-context inference	Low per-task tuning burden; designed specifically for small tabular settings	Domain coverage and scaling assumptions; should be compared under the same information and evaluation rules

The key limitation is search-space dependence. An AutoML system cannot discover a model or transformation that its search space does not contain. Therefore, performance is a function of the dataset, search space, optimizer, and computational budget, not the optimizer alone.

3.3 Agentic Machine Learning

Agentic systems broaden the optimization target. Instead of selecting only a configuration λ , an agent may alter preprocessing, feature engineering, model choice, evaluation code, and experimental strategy. AutoML-Agent demonstrates this direction using specialized LLM agents, retrieval-augmented planning, parallel sub-task execution, and multi-stage verification [?].

The additional flexibility introduces additional failure modes. Generated code may contain invalid assumptions, leakage, unnecessary complexity, or unstable decisions. An agent can also repeatedly inspect validation results and revise its pipeline, creating a meta-level optimization loop. Thus, successful pipeline generation is not sufficient evidence of statistically reliable generalization.

3.4 Small-Data Specialized Models

TabPFN represents a different strategy: rather than searching extensively for every new dataset, a pretrained model embeds prior structure learned from synthetic tabular problems [?]. The literature therefore suggests that small-data benchmarking should include specialized models rather than treating conventional AutoML and expert-tuned classical models as the only alternatives.

4 Literature Synthesis

Table 1 summarizes the four families using the dimensions required for a meaningful review.

4.1 AutoML and Benchmark Protocols

AMLB provides a particularly relevant methodological reference because it evaluates multiple AutoML frameworks under standardized tasks, resource allocation, and multifaceted outcome measures [?]. Its results demonstrate why a benchmark should not reduce comparison to a single

final accuracy number. Framework failures and inference-time trade-offs can affect the practical interpretation of performance.

For small datasets, this concern becomes stronger. A small validation set can create high-variance estimates, and repeated configuration evaluation can indirectly overfit that validation set. Consequently, benchmark conclusions should be interpreted jointly with the validation design and search budget.

4.2 Small-Data Models and the Role of Priors

The TabPFN literature provides evidence for a different design philosophy. TabPFN uses prior-data fitted learning and in-context prediction rather than conventional per-dataset hyperparameter optimization [?]. The relevant review implication is not that a specialized model should automatically replace AutoML, but that the benchmark must include models whose computational strategy is fundamentally different from search-based optimization.

The comparison should therefore distinguish between “search efficiency” and “model prior efficiency.” A system can require fewer experiments because it has a strong pretrained prior, not because it searches a configuration space more efficiently.

4.3 Agentic Automation and Evaluation

AutoML-Agent shows that LLM-based agents can coordinate multiple stages of an ML workflow, including planning, preprocessing, model design, execution, and verification [?]. This expands what counts as automation. However, the literature also implies that agentic systems introduce additional experimental variables: LLM version, temperature, prompt, retrieved information, tool availability, generated code, and external information.

These variables should be logged and controlled where possible. Otherwise, a benchmark may compare not only modeling paradigms but also unequal information environments.

5 Comparative Evaluation Framework

5.1 Data Splitting and Leakage Control

The test set should remain completely inaccessible to all model-selection procedures. For small datasets, nested cross-validation is preferable: outer folds estimate generalization while inner procedures perform model selection. All preprocessing transformations must be fitted inside the relevant training folds.

For agentic systems, leakage control must additionally cover generated searches, documentation retrieval, and external information. If internet access is permitted, the same information-access policy should apply to every system or be treated as an explicit experimental factor.

5.2 Predictive Performance

Accuracy alone is insufficient. For classification, a review-grounded benchmark should consider accuracy, balanced accuracy, macro-F1, ROC-AUC where appropriate, PR-AUC for imbalanced problems, log loss, Brier score, and calibration error. For regression, RMSE, MAE, R^2 , and domain-specific metrics can be reported.

A composite score can summarize results, but primary analysis should retain individual metrics because aggregation can conceal trade-offs.

5.3 Computational Cost

Predictive performance should be interpreted jointly with computational cost. An illustrative efficiency quantity is

$$\text{Efficiency}_i = \frac{\text{Performance}_i}{\text{Cost}_i}, \quad (3)$$

where cost can include wall-clock time, CPU/GPU hours, memory, number of model evaluations, monetary API cost, and energy where measurable.

The quantity in Equation 3 is a reporting aid rather than a universal objective. Different studies may weight cost components differently, so the underlying measurements should always be reported separately.

5.4 Human Effort

Human involvement should be explicitly measured. A simple operationalization is

$$H = T_{\text{setup}} + T_{\text{debug}} + T_{\text{analysis}} + T_{\text{intervention}}. \quad (4)$$

This allows a benchmark to distinguish computational automation from human-effort reduction.

5.5 Stability and Reproducibility

Let $M_{i,s}$ denote the performance of method i under seed or resampling condition s . A benchmark should report both the mean and standard deviation across repeated conditions. This is important because small datasets can produce substantial changes in model rankings across random splits.

Agentic systems require additional reproducibility records: model version, temperature, prompt, tools, retrieved documents, generated code, execution trace, and API version where applicable.

6 Threats to Validity in Existing Comparisons

6.1 Validation Overfitting

If K candidate configurations are evaluated on the same limited validation set, the selected configuration is the minimum of K noisy estimates. Even when configurations have similar true risk, the selected validation score can be optimistically biased. This affects experts, AutoML systems, and agents alike.

6.2 Search-Space Bias

A system that cannot represent a model family cannot discover it. Consequently, claims about the superiority of an AutoML system are incomplete unless the search space, preprocessing operators, ensemble options, and computational budget are documented.

6.3 Agent Nondeterminism

An LLM agent may generate different workflows from the same dataset and task description. Independent runs are therefore necessary. Success rate, predictive performance conditional on successful completion, and failure modes should be reported separately.

6.4 Unequal Information Access

An agent with internet access, retrieved documentation, or external datasets may have information unavailable to a conventional baseline. Such differences can produce an apparent modeling advantage that is actually an information-access advantage.

6.5 Unequal Human Intervention

An expert baseline may receive hours of manual tuning while an automated baseline receives a fixed compute budget. Alternatively, an agent may receive repeated human corrections. Both situations make direct comparison difficult unless intervention rules are explicitly recorded.

7 Research Gaps

7.1 Gap 1: Direct Expert–AutoML–Agent Comparisons

Most established AutoML benchmarks compare automated frameworks with each other. AMLB is an important controlled benchmark, but its primary objective is framework-level comparison rather than a unified expert-versus-AutoML-versus-agent study [?]. A small-data benchmark should therefore include a strong, protocolized expert baseline.

7.2 Gap 2: Dedicated Small-Data Evaluation

Many benchmark suites mix datasets across a wide range of sizes. Treating small datasets as one subset can obscure failure modes caused by high validation variance. A dedicated benchmark around the operational criterion in Equation 1 would permit more focused analysis.

7.3 Gap 3: Agentic Success Versus Generalization

A successful agent run demonstrates that a workflow can be generated and executed. It does not by itself demonstrate better out-of-sample generalization. The distinction between automation success and statistical superiority should therefore be explicit.

7.4 Gap 4: Human Expertise Is Poorly Operationalized

“Expert” is not a standardized treatment. Experts differ in domain knowledge, experience, preferred algorithms, and available time. Future studies should specify expert protocols and record human effort using measures such as Equation 4.

7.5 Gap 5: Cost–Performance Trade-offs

An agent may obtain a small predictive improvement while consuming substantially more computation or API calls. Conversely, a specialized small-data model may require little per-dataset optimization. Cost-performance analysis is therefore necessary for practical conclusions.

8 Proposed Benchmark Protocol

Figure 2 summarizes a protocol derived from the methodological issues identified in the review.

The protocol has six important controls. First, datasets must be characterized before model selection. Second, the test set must be isolated. Third, compute and information-access budgets should be documented. Fourth, all four methodological families should follow the same evaluation boundary. Fifth, repeated runs should be used to estimate stability. Sixth, conclusions should be based on multiple outcome dimensions rather than a single score.

9 Discussion

The literature does not support a meaningful universal ordering of expert tuning, AutoML, agentic ML, and specialized small-data models. Instead, each family optimizes a different part of the problem.

Expert tuning provides flexible inductive reasoning but is constrained by human time and bias. Conventional AutoML provides systematic search but is constrained by the search space and the quality of the validation signal. Agentic systems provide semantic workflow adaptation but add nondeterminism and a larger reproducibility surface. Specialized models reduce per-task

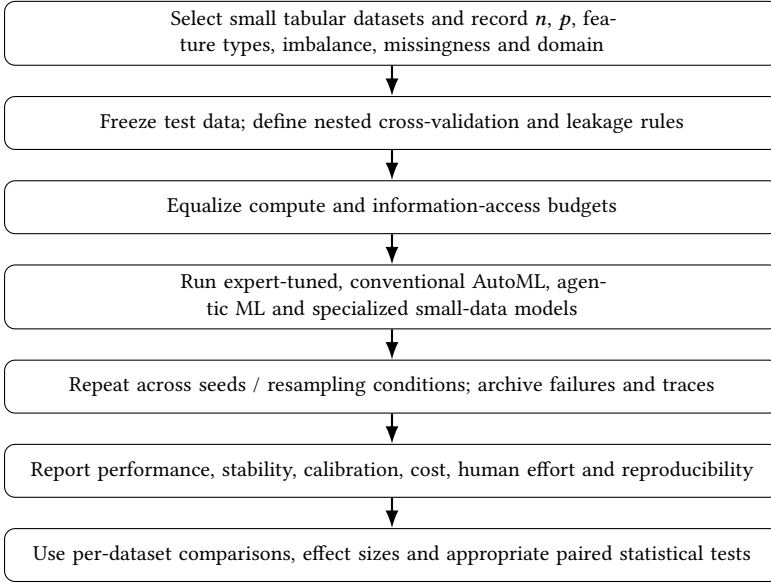


Fig. 2. Proposed controlled benchmarking workflow derived from the reviewed methodological gaps.

Table 2. Recommended benchmark variables and controls

Dimension	Minimum control	Reason
Dataset	Record n , p , feature types, imbalance, missingness and domain	Allows results to be interpreted by dataset characteristics
Validation	Nested cross-validation; test set hidden	Reduces selection bias and prevents test-set optimization
Search budget	Fixed wall-clock or compute budget	Prevents resource imbalance
Information access	Same external-information policy	Separates modeling capability from information advantage
Randomness	Multiple seeds / resampling runs	Measures stability and nondeterminism
Human effort	Record setup, debugging, analysis and intervention time	Makes automation claims measurable
Agent trace	Store prompts, model version, tools and generated code where permitted	Supports reproducibility and auditing
Metrics	Performance, stability, calibration, cost and failure rate	Avoids single-metric conclusions

optimization by embedding prior structure, but their usefulness depends on how well that prior matches the target regime.

This framing also changes how “performance” should be interpreted. A method with a slightly higher predictive score but substantially higher compute, API cost, or human intervention may not provide the same practical value as a method with comparable predictive performance at lower

cost. Likewise, a method with a marginally lower mean score but much lower variance may be preferable for a deployment-sensitive setting.

The review therefore suggests that future empirical work should report a profile rather than a single winner. Per-dataset results, variance, calibration, failures, compute, and human effort should accompany aggregate comparisons. Mean rank can be useful as a summary:

$$\bar{r}_j = \frac{1}{N} \sum_{i=1}^N r_{ij}, \quad (5)$$

but it should not replace dataset-level analysis.

10 Future Research Directions

10.1 Human-in-the-Loop AutoML

A promising direction is adaptive collaboration rather than full automation. A hybrid workflow could be represented as Human $\rightarrow$ Agent $\rightarrow$ AutoML $\rightarrow$ Human, with experts providing high-level constraints while automated systems perform detailed search.

10.2 Meta-Learning for Small Datasets

Future AutoML systems could exploit information from previous small-data tasks to form priors over promising configurations before search. This could reduce repeated exploration when related datasets recur.

10.3 Agentic Statistical Validation

LLM agents should become stronger not only at code generation but also at statistical validation. Useful capabilities include automatic detection of leakage, insufficient validation samples, unstable metrics, class imbalance, selection bias, multiple-testing effects, and calibration failure.

10.4 Confidence-Aware Automation

A useful future system should return more than a selected model. It should report estimated performance, uncertainty, stability, cost, failure probability, and the rationale for selection. Such outputs would make automated systems easier to audit.

10.5 Anytime Benchmarking

Methods should be evaluated at multiple budgets, for example 1, 5, 15, 60, and 240 minutes. This provides information about how quickly a method reaches useful performance rather than only its final state. Resource-aware methods such as Hyperband, BOHB, and FLAML provide methodological motivation for this direction [? ? ?].

11 Conclusion

AutoML has changed the economics of machine-learning development by automating model selection, hyperparameter optimization, preprocessing, and ensemble construction. The literature demonstrates several successful search strategies, from Bayesian optimization and genetic programming to bandit-based and cost-aware approaches [? ? ? ?]. Benchmarking work also demonstrates that framework comparisons depend strongly on evaluation protocol [?].

Small datasets introduce a distinctive statistical challenge because validation noise and model-selection variance can dominate the apparent benefit of larger automated search budgets. Specialized models such as TabPFN show that an alternative to repeated per-dataset search is possible [?].

LLM-based agents extend automation further by allowing systems to plan, execute, inspect, and revise workflows [?].

The central methodological conclusion of this review is that these approaches should not be compared using predictive accuracy alone. A defensible benchmark must control validation, compute budgets, information access, leakage, randomness, and human intervention, while reporting stability, calibration, cost, human effort, reproducibility, and failure modes. The research gap is therefore not simply to determine which paradigm is “best”, but to identify the data and resource regimes in which each paradigm provides a meaningful advantage.

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